Tool Retry Impact

Does it really make LLM agents that much better?

- Kani detects when LLMs hallucinate or miss arguments to tools and gives the LLM detailed feedback

- The LLM can retry the tool call, and recover from simple errors

- We test open-weight LLMs of differing sizes on the Berkeley Function Calling Leaderboard (BFCL)
  - **Llama 3**: 1B, 3B, 8B
  - **Qwen 3**: 0.6B, 1.7B, 4B, 8B, 14B, 30B
  - **GPT-OSS**: 20B, 120B

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<|im_start|>system
You can use the following functions:
get_weather(location: str)<|im_end|>

```
<|im_start|>assistant
<tool_call>
{"name": "get_weather", "params": {
  "location": "Philadelphia, PA",
  "unit": "f"
}}
</tool_call><|im_end|>
```

```
<|im_start|>function
1 validation error in get_weather
  unit
    unexpected argument<|im_end|>
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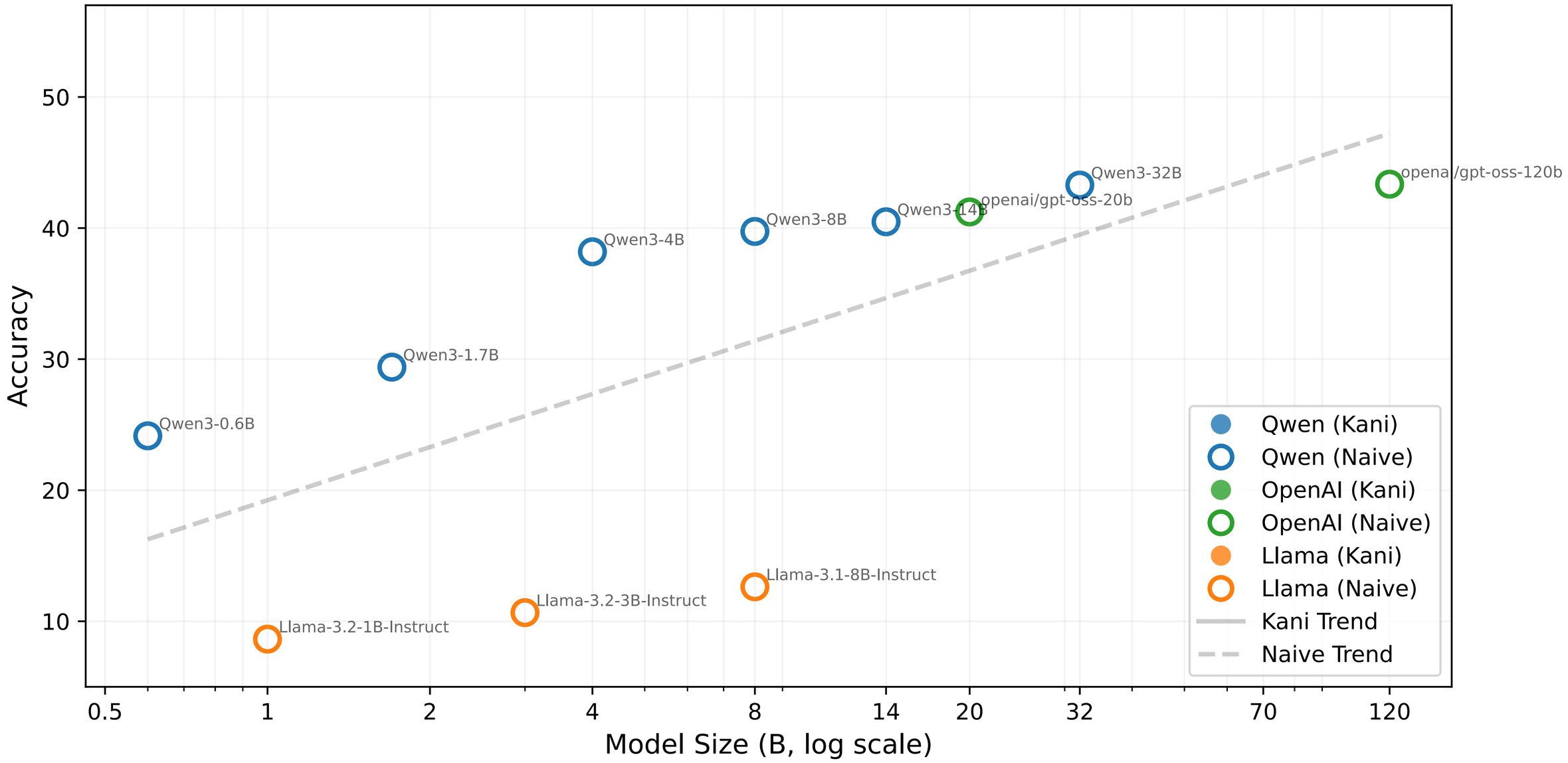
Accuracy
50
40
30
20
10
0.5
1
2
4
8
14
20
32
70
120
Model Size (B, log scale)
Qwen3-0.6B
Qwen3-1.7B
Qwen3-4B
Qwen3-8B
Qwen3-14B
Qwen3-32B
openai/gpt-oss-20b
opena /gpt-oss-120b
Llama-3.2-1B-Instruct
Llama-3.2-3B-Instruct
Llama-3.1-8B-Instruct
Qwen (Kani)
Qwen (Naive)
OpenAI (Kani)
OpenAI (Naive)
Llama (Kani)
Llama (Naive)
Kani Trend
Naive Trend